ACADEMIUM ENGINE PACKET

Section 4: KPI Computation

Version: 2.0 **Status:** Active Specification **Scope:** Applies to all KPI metric movements across every turn in every case, regardless of step count (4–10 turns) or decision format (MCQ or FR). **Runs in parallel with:** Section 3 (Consequence Generation). These two sections receive the same inputs and operate independently. Neither determines the other. **Inputs received from:** Section 2 (extracted signals, RDS band), Turn 1 tradeoff signature, professor's KPI configuration, accumulated KPI state from prior turns, professor's tradeoff configuration **Outputs sent to:** Section 6 (Causal Explanation Generation — receives KPI movement data to generate "Why did this change?" content), Section 7 (Response Assembly — receives final KPI delta package for display)

4.1 Purpose and Pedagogical Foundation

KPI movements are the simulation's feedback mechanism — the system of signals that tells students their decisions had real, measurable effects on the organization they are navigating. They are not grades. They are not scores. They are not rewards or punishments. They are data from the world the student is operating in.

The engine's job in this section is to compute which metrics moved, in which direction, and by how much — based entirely on what the student decided and what the case's directional logic says that decision produces. The computation must be consistent, traceable, and honest. A student should never look at a KPI movement and wonder why it happened. Every movement must be explainable in plain language that directly references their decision.

Why KPIs Run in Parallel with Consequences

Sections 3 and 4 both receive the same inputs but operate independently. This is a deliberate architectural choice grounded in a critical principle: **the narrative does not determine the metrics, and the metrics do not determine the narrative.** Both emerge from the same source — the student's decision signals and the case's directional logic — independently, and are then assembled together in Section 7.

If the narrative determined the metrics, the engine would be writing a story and then generating numbers to match, which is reverse-engineered and fragile. If the metrics determined the narrative, the engine would be illustrating a scorecard, which violates the pedagogical posture. The parallel architecture prevents both failure modes.

Research Basis

Cognitive Load Theory (Sweller, 1988): Displaying too many metrics simultaneously overwhelms working memory. The engine limits visible KPIs to 2–3 per turn and only surfaces metrics that have actually moved. This keeps the student's cognitive resources focused on the consequence, not on parsing a dashboard.

Feedback Intervention Theory (Kluger & DeNisi, 1996): Metric feedback must be task-directed, not ego-directed. KPI movements show what changed in the organization — they do not evaluate the student's performance. The "Why did this change?" affordance ensures every movement is anchored to the decision that caused it, not to an abstract judgment.

Self-Determination Theory (Deci & Ryan, 1985): Students must feel that their choices had genuine, differentiated effects. If all decisions produce the same KPI movements, the simulation has failed the autonomy requirement. Metric computation must be sensitive to the content of the student's specific decision — not just its presence.

Experiential Learning (Kolb, 1984): KPI movements are the quantitative layer of the Reflective Observation stage. They give the student a structured way to observe the organizational impact of their decision alongside the narrative. The combination of story + signal is more powerful than either alone.

4.2 The Five Canonical KPIs

These are the only KPIs in the Academium system. No case may introduce a KPI outside this set. Professors select 3–5 of these for each case they configure.

KPI ID	Name (English)	Name (Spanish)	What It Represents
K1	Budget / Financial Impact	Presupuesto / Impacto Financiero	Financial health, cost management, revenue trajectory, resource allocation efficiency
K2	Team Morale	Moral del Equipo	Internal team sentiment, willingness to collaborate, psychological safety within the organization, retention risk
K3	Brand Reputation	Reputación de Marca	External perception by clients, partners, and market; organizational credibility; trust from outside the organization
K4	Operational Efficiency	Eficiencia Operativa	Process quality, delivery reliability, timeline adherence, resource utilization, execution capacity
K5	Stakeholder Trust	Confianza de Stakeholders	Trust level among key internal and external stakeholders — investors, clients, board members, regulators, partners

KPI Domain Definitions

Each KPI has a defined domain — the organizational territory it covers. The engine must only move a KPI when the student's decision has a clear causal connection to that KPI's domain. Moving a KPI for which no causal connection exists is a fabrication and a pipeline failure.

KPI	Domain — What Decisions Affect It
K1	Decisions involving resource allocation, cost management, revenue risk, financial commitments, budget trade-offs
K2	Decisions affecting the internal team's workload, recognition, autonomy, trust in leadership, work conditions
K3	Decisions that affect how the organization is perceived externally — client-facing communication, public commitments, market behavior
K4	Decisions affecting how work gets done — process changes, timeline adjustments, capacity allocation, delivery commitments
K5	Decisions affecting the confidence of specific named stakeholders (investors, board, key clients, regulatory bodies, strategic partners)

4.3 Professor KPI Configuration

The professor selects 3–5 active KPIs when configuring a case. The engine must:

- Only compute movements for active KPIs
- Never surface an inactive KPI to the student
- Never reference an inactive KPI in a causal explanation

This is enforced at the start of every session. The active KPI set is locked for the duration of that case deployment and cannot change mid-simulation.

Configuration Validation Rule

If a professor has configured only 3 active KPIs and all three are in closely related domains (e.g., K1, K4, and K4-adjacent), the case review process should flag this as a potential design issue — limited KPI diversity reduces the range of trade-offs the simulation can produce. However, the engine does not reject this configuration; it operates within whatever the professor has defined.

4.4 KPI Direction Logic

Direction determines whether a KPI moves up (positive) or down (negative) following a decision. Direction logic differs between MCQ turns and FR turns.

Turn 1 (MCQ): Pre-Authored Direction

For the first turn, direction is entirely pre-authored. Each MCQ option carries a tradeoff signature that explicitly encodes the direction of every affected KPI:

```
Example Tradeoff Signature (Option A):
{
  K1: -1,      // Budget decreases (small)
  K2: +1,      // Team Morale increases (small)
  K3: +1,      // Brand Reputation increases (small)
  K4:  0,      // Operational Efficiency unaffected
  K5: +2       // Stakeholder Trust increases (moderate)
}
```

The engine applies this signature directly. Direction is not inferred; it is read from the pre-authored data. This is the most reliable turn in the simulation because there is no ambiguity in

the direction computation.

FR Turns (Intermediate and Final): Signal-Anchored Direction

For FR turns, direction is inferred through a two-layer process:

Layer 1: Case Directional Architecture (Hard Constraints)

Every case has a pre-authored directional architecture — a set of fundamental trade-off relationships that define the organizational dynamics of the scenario. These constraints cannot be overridden by student signals.

Examples:

- *"In this case, decisions that prioritize client accommodation always create at least some pressure on K1 (Budget) or K4 (Operational Efficiency)."*
- *"In this case, decisions that prioritize internal team protection always create at least some tension with K5 (Stakeholder Trust) from the client side."*

These are authored by the case designer and encoded in the case definition. They define which directions are possible, not which direction will occur for any given student.

Layer 2: Signal-Based Direction Selection (Within Constraints)

Within the constraints defined by Layer 1, the student's extracted signals determine the specific direction of movement for each active KPI. The engine reads signal content — not just signal presence — to make this determination.

Student Signal Content	Direction Inference
Intent signals a priority that directly benefits a KPI domain	That KPI moves in a positive direction
Tradeoff Awareness explicitly names a KPI domain as a cost	That KPI moves in a negative direction
Stakeholder Awareness names a stakeholder associated with a KPI	That KPI is flagged as affected; direction depends on whether the stakeholder's interests were served or strained
Ethical Awareness surfaces a transparency or fairness concern	K3 (Brand Reputation) and K5 (Stakeholder Trust) are flagged; direction depends on whether the ethical concern was addressed or ignored
No signal present for a KPI domain	That KPI is not affected this turn (unless a pre-authored Tier 3 event is triggered)

Direction Conflict Resolution

If signal content suggests a KPI should move positively but the case directional architecture constrains it to move negatively, the constraint wins. The engine does not override case architecture based on signal quality. This preserves the integrity of the simulation's organizational logic.

However: if a student's signals are exceptionally strong and multi-dimensional (Integrated RDS band), the engine may reduce the magnitude of a constrained negative movement (e.g., from Tier 2 to Tier 1) to reflect that a more thoughtful decision produced a less severe negative consequence. Direction cannot be reversed; magnitude can be attenuated.

4.5 Tier Model: Movement Magnitude

Every KPI movement is assigned to one of three tiers that define its magnitude. Tiers are displayed to students using qualitative labels, never numerical values.

Tier	Internal Label	Student-Facing Label	Frequency Target
Tier 1	Small	*Ligero / Slight*	60–70% of all movements
Tier 2	Moderate	*Moderado / Moderate*	25–35% of all movements
Tier 3	Significant	*Significativo / Significant*	5–10% of all movements

Tier Assignment Rules

Tier 3 — Strict Pre-Authoring Requirement Tier 3 movements are never generated autonomously by the engine. They are pre-authored in the case definition as specific trigger events. The engine checks whether the student's decision activates a pre-authored Tier 3 trigger. If yes, Tier 3 applies. If no pre-authored trigger exists for this decision, Tier 3 cannot be assigned regardless of signal quality or RDS band.

This rule exists because Tier 3 movements represent significant organizational events — a major client relationship collapse, a budget crisis, a team walkout, a reputational incident. These must be deliberate case design choices, not generative outputs.

Tier 2 — Signal Quality + RDS Band Required Tier 2 is available for Intermediate and Integrated RDS band responses. Assignment requires:

- The student's signal for the relevant KPI domain must be PRESENT or STRONG quality

- The case directional architecture must permit a Tier 2 movement in this domain for this turn
- At least one prior turn must have established context in this domain (first-time KPI movements are typically Tier 1)

Tier 1 — Default All movements default to Tier 1 unless Tier 2 or Tier 3 conditions are met. Surface band RDS responses always produce Tier 1 movements only.

Tier Attenuation Rule

When the student's reasoning demonstrates strong Tradeoff Awareness for a domain where a negative movement is pre-architectured, the engine may attenuate the tier by one level (Tier 2 → Tier 1). This reflects the real-world principle that anticipating a cost and planning for it reduces its impact. Direction does not change. Tier may reduce by one level maximum.

4.6 KPI Selection: Which Metrics to Display Per Turn

Even if multiple active KPIs are affected by a decision, the engine displays a maximum of 3 KPIs per turn. This constraint is pedagogical, not cosmetic — cognitive load limits require focus.

Selection Priority Rules

When more than 3 active KPIs are affected, the engine selects the 3 to display using this priority order:

1. **Highest tier movement first.** A Tier 2 movement is always displayed over a Tier 1 movement.
2. **First-appearance KPIs second.** If a KPI is appearing for the first time this session, it takes priority over a KPI that has already appeared in prior turns. New information is more valuable to the student than additional data on something already established.
3. **Highest signal-quality alignment third.** If the student's response contained a strong or present signal for a specific KPI domain, that KPI takes priority over one where the connection is inferred rather than signaled.
4. **Narrative alignment fourth.** The KPIs whose movements are most explicitly referenced in the Section 3 consequence narrative take priority. The displayed metrics and the narrative must be in agreement — if the narrative mentions team pressure, Team Morale should be visible.

Never-Display Rule

A KPI that was not affected this turn — meaning it received no directional signal and no pre-authored trigger — must never be displayed, even if showing it would make the dashboard look more complete. Displaying an unmoved KPI implies the decision affected it when it did not. This is a fabrication and a trust violation.

4.7 KPI Accumulation Model: 4–10 Turn Cases

KPIs accumulate across all turns in the simulation. The engine maintains an internal KPI state object that tracks the cumulative history of every active KPI. This state is used to inform tier decisions, consequence richness (Section 3), and the Final Outcome narrative.

Accumulation State Object

For each active KPI, the engine maintains:

```
KPI State {
  kpi_id: K1–K5,
  current_trajectory: "positive" | "negative" | "mixed" | "neutral",
  total_turns_moved: integer,
  consecutive_negative_turns: integer,
  consecutive_positive_turns: integer,
  last_tier: 1 | 2 | 3,
  cumulative_delta: integer  // internal only; never displayed as a number
}
```

Accumulation Rules

Rule 1: Trajectory classification After each turn, the engine classifies the KPI's trajectory based on cumulative movement pattern:

- *Positive*: more positive movements than negative across the session
- *Negative*: more negative movements than positive
- *Mixed*: roughly equal positive and negative movements
- *Neutral*: KPI has not moved or has returned to approximate baseline

Rule 2: Consecutive stress escalation If a KPI has moved negatively for 3 or more consecutive turns, the next movement in that KPI domain carries elevated narrative weight in Section 3 — even if the tier remains Tier 1. The consequence narrative must acknowledge the accumulated pressure, not treat it as a fresh event.

Rule 3: Tier escalation after sustained stress If a KPI has been in negative trajectory for 4+ turns AND the current turn's signal for that domain is PRESENT or STRONG, the engine may assign Tier 2 even if the standard Tier 2 conditions would not otherwise be met. This reflects the organizational reality that sustained stress compounds into bigger consequences over time.

Rule 4: Recovery attenuation If a KPI has been in negative trajectory and the student's current decision produces a positive movement in that domain, the positive movement is always Tier 1 — regardless of signal quality. Recovery from sustained organizational damage is slow and incremental. A single good decision does not reverse accumulated stress immediately.

Rule 5: Final turn synthesis At the Final turn, the engine computes the cumulative trajectory for each active KPI across the full session. This synthesis data is passed to the Final Outcome narrative generator (case structure Section 8), not displayed as a scorecard. The trajectory informs the tone and content of the closing narrative — a student with a consistently positive K5 trajectory should see a different Final Outcome than one with a collapsed K5 — but neither trajectory is labeled as "better."

4.8 Anti-Patterns: What the Engine Must Prevent

The following outcomes indicate a failure in KPI computation logic. The engine must detect and correct these before output is delivered.

Anti-Pattern 1: Uniform Direction (All KPIs Moving the Same Way)

What it looks like: Every KPI that moves this turn moves in the same direction — all positive or all negative.

Why it's a failure: A decision with no trade-offs is not a business decision. If every metric improves, the student has been told their decision was universally good. If every metric declines, the student has been punished. Both violate the Tradeoff Posture.

Correction rule: When tradeoffs are configured (States A or B from Section 3.6), at least one active KPI must move in the opposite direction of the majority. The engine must review the full movement set before output and enforce this. If all computed movements are in the same direction, the engine selects the KPI with the weakest signal alignment and flips its direction to Tier 1 in the opposite direction, consistent with case directional architecture.

Exception: When tradeoffs are not configured (State C), uniform direction is permitted — but the engine should still check whether the case's directional architecture makes this realistic.

Anti-Pattern 2: Zero Movement

What it looks like: No active KPI moves following a student decision.

Why it's a failure: A decision with zero organizational impact communicates to the student that their choice did not matter. This destroys engagement and undermines the simulation's entire value proposition.

Correction rule: Every submitted decision must produce at least one active KPI movement. If signal extraction produced no domain-relevant signals for any active KPI, the engine selects the KPI most aligned with the student's stated intent and applies a Tier 1 movement in the direction consistent with case architecture.

Anti-Pattern 3: Movement Without Narrative Support

What it looks like: A KPI moves, but the Section 3 consequence narrative contains no reference to the domain that KPI represents.

Why it's a failure: If Team Morale drops but the narrative says nothing about the team, the student sees a number change with no story attached. This feels arbitrary and punitive.

Correction rule: This is a cross-section consistency check in Section 7 (Response Assembly). However, the KPI computation step must flag any movement where the corresponding narrative domain is absent so that Section 7 can catch it. If the flag is raised and narrative support cannot be added, the KPI movement is removed rather than displayed without support.

Anti-Pattern 4: Tier 3 Without Pre-Authored Trigger

What it looks like: The engine generates a Tier 3 movement based on signal quality or RDS band alone, without a pre-authored case trigger.

Why it's a failure: Tier 3 events are significant organizational crises. Generating them autonomously introduces unpredictability and can produce consequences that feel punitive or arbitrary. Tier 3 must always be a deliberate case design choice.

Correction rule: Any Tier 3 movement without a traceable pre-authored trigger is automatically downgraded to Tier 2 before output.

Anti-Pattern 5: KPI Stagnation Across Full Case

What it looks like: An active KPI never moves across the entire simulation, despite being configured as active.

Why it's a failure: A configured KPI that never moves was either a poor case design choice (the KPI was irrelevant to the scenario) or the engine failed to detect relevant signals. Either way, the dashboard will show a KPI with no history, which creates confusion.

Correction rule: This is a case design validation issue, not an engine runtime issue. The case review process (pre-deployment) should flag any case where a configured KPI has no designed movement triggers. At runtime, if a configured KPI has not moved by the second-to-last turn, the

engine checks whether the current turn's signals provide any valid connection to that KPI domain and surfaces a Tier 1 movement if one can be justified.

4.9 Display Rules: What the Student Sees

Directional Indicators

KPI movements are displayed as directional arrows with magnitude labels. No numbers. No percentages. No "out of 10" scales.

Display Element	Specification
Direction	Up arrow (↑) for positive movement, Down arrow (↓) for negative movement
Magnitude	Text label only: *Ligero/Slight* (Tier 1), *Moderado/Moderate* (Tier 2), *Significativo/Significant* (Tier 3)
KPI label	The KPI's contextual name for this case (not the generic canonical name — see 4.9.1)
"Why?" affordance	Visible on every movement, every turn, without exception

4.9.1 Contextual KPI Labeling

KPIs must be labeled using language that is specific to the case's organizational context, not the generic canonical name. This keeps the metrics feeling like organizational signals, not abstract academic categories.

Canonical Name	Example Contextual Label (Case 001)
Budget / Financial Impact	Salud Financiera del Proyecto / Project Financial Health
Team Morale	Cohesión del Equipo / Team Cohesion
Brand Reputation	Percepción del Cliente / Client Perception
Operational Efficiency	Capacidad de Entrega / Delivery Capacity
Stakeholder Trust	Confianza de Partes Clave / Key Stakeholder Confidence

4.9.2 First Appearance Rule

A KPI is not displayed until it has moved at least once. The metric panel does not show all active KPIs from the start of the simulation — it populates progressively as KPIs become relevant to the decisions being made. This reduces initial cognitive load and makes each new KPI appearance feel meaningful rather than routine.

4.9.3 The "Why Did This Change?" Affordance

Every displayed KPI movement must have an accessible "Why did this change?" explanation. This affordance is mandatory without exception. It is generated by Section 6 (Causal Explanation Generation) using the KPI movement data produced in this section. The explanation must:

- Reference the student's specific decision (not a generic organizational statement)
- Explain the causal chain from decision to metric movement in plain language
- Be 2–4 sentences maximum
- Never use prohibited language from Section 3.7
- Never imply the movement was avoidable through a "better" decision

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4.10 Quality Gates

Before any KPI movement set is passed to Section 6 and Section 7, it must pass all six quality gates.

Gate	Check	Pass Condition
G1: Minimum movement	Did at least one active KPI move?	Yes
G2: Tradeoff compliance	When tradeoffs are configured, does at least one KPI move opposite the majority?	Yes (or N/A if State C)
G3: Tier 3 authorization	Are all Tier 3 movements traceable to a pre-authored trigger?	Yes — no autonomous Tier 3
G4: Display limit	Are 3 or fewer KPIs selected for display?	Yes
G5: Domain-signal traceability	Can every moving KPI be traced to either a student signal or a pre-authored trigger?	Yes — no fabricated movements
G6: Narrative alignment flag	Are any moving KPIs in domains not referenced in the Section 3 narrative?	All flagged for Section 7 cross-check

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4.11 Worked Examples

All examples assume Case 001 (Spanish, international business/operations). Active KPIs: K1, K2, K4, K5.

EXAMPLE A — Turn 1, MCQ, Pre-Authored Tradeoff Signature

Student selected: Option B — *"Priorizar la relación con el cliente, siendo transparentes sobre las limitaciones operativas"*

Pre-authored tradeoff signature:

```
{K1: -1, K2: +1, K4: -1, K5: +2}
```

KPI Computation Output:

KPI	Direction	Tier	Display?
K1 (Financial)	↓ Negative	Tier 1	Yes (first appearance)
K2 (Team Morale)	↑ Positive	Tier 1	Yes (first appearance)
K4 (Op. Efficiency)	↓ Negative	Tier 1	Yes (first appearance)
K5 (Stakeholder Trust)	↑ Positive	Tier 2	Yes (highest tier; first appearance)

Gate checks: G1 ✓ G2 ✓ (K1, K4 negative; K2, K5 positive) G3 ✓ (no Tier 3) G4 ✓ (4 active but engine selects top 3: K5, K1, K2 — K4 deferred to next turn when it moves again or narrative supports it) G5 ✓ G6 ✓

EXAMPLE B — Turn 4 of 8, FR, Surface Band (RDS 3)

Accumulated state entering Turn 4: K1 negative trajectory (2 consecutive negative turns), K5 positive trajectory, K2 neutral.

Student input: *"Hablaría con el equipo."* (Surface band — intent present, all others absent)

Direction computation:

- K2 (Team Morale): Intent toward team communication → positive signal → K2 ↑ Tier 1
- K1 (Financial): No signal in financial domain → no movement
- K4 (Op. Efficiency): No signal → no movement
- K5 (Stakeholder Trust): No signal → no movement

KPI Computation Output:

KPI	Direction	Tier	Notes
K2 (Team Morale)	↑ Positive	Tier 1	Only movement this turn

Gate checks: G1 ✓ G2 — only one KPI moved; tradeoff check: case architecture requires at least one cost. Engine applies correction: K4 receives Tier 1 negative movement (team communication consumes delivery capacity time — consistent with case architecture). G3 ✓ G4 ✓ G5 ✓ G6 — flag for Section 7 to ensure narrative references both K2 and K4 domains.

EXAMPLE C — Turn 4 of 8, FR, Integrated Band (RDS 14) — Same Turn, Richer Response

Same accumulated state as Example B.

Student input: Multi-signal response demonstrating all five signals at PRESENT/STRONG. Intent: restructure team communication protocols. Tradeoff: delivery timeline will slip. Stakeholder: operations director named. Ethical: transparency with client about adjusted capacity.

Direction computation:

- K2 (Team Morale): Strong stakeholder signal (operations director) + intent toward team → ↑ Tier 2 (consecutive neutral turns + strong signal qualifies)
- K4 (Op. Efficiency): Tradeoff Awareness explicitly names delivery timeline → ↓ Tier 1 (constrained by architecture)
- K5 (Stakeholder Trust): Ethical Awareness around client transparency → ↑ Tier 1 (client trust benefits from transparency)
- K1 (Financial): No direct signal; architecture doesn't trigger → no movement

KPI Computation Output:

KPI	Direction	Tier	Display?
K2 (Team Morale)	↑ Positive	Tier 2	Yes (highest tier)
K4 (Op. Efficiency)	↓ Negative	Tier 1	Yes (tradeoff compliance)
K5 (Stakeholder Trust)	↑ Positive	Tier 1	Yes (first appearance this direction)

Comparison with Example B: Same turn, same case, different student reasoning. The Integrated band student sees a more differentiated, nuanced metric picture — not a "better" one. K2 moved at Tier 2 instead of Tier 1 because the richer reasoning demonstrated a specific, defensible action with stronger organizational impact. K5 appeared because ethical awareness surfaced a stakeholder dimension. K4 moved negatively in both cases — the trade-off was real regardless of reasoning quality.

4.12 Section Summary: What the Engine Must Do

At every decision turn, the KPI computation engine must:

1. **Run in parallel with Section 3** — receive the same inputs but operate independently. Never derive KPI movements from the consequence narrative, and never derive the narrative from KPI movements.
2. **Check the professor's active KPI configuration** — only compute movements for configured KPIs.
3. **Apply direction logic** — Layer 1 (case architecture constraints) then Layer 2 (signal-based selection within constraints).
4. **Assign tiers** — Tier 1 default, Tier 2 when conditions met, Tier 3 only from pre-authored triggers.
5. **Apply accumulation state** — check prior turn history, apply consecutive stress escalation and recovery attenuation rules.
6. **Select display KPIs** — maximum 3, using the four-level priority order.
7. **Check all six quality gates** — correct anti-patterns before passing output downstream.
8. **Flag narrative alignment issues** — identify any moving KPI whose domain is absent from the Section 3 narrative, for resolution in Section 7.
9. **Pass the full KPI movement package to Sections 6 and 7** — including tier, direction, domain, and the signal content that drove each movement, so causal explanations can be generated accurately.

Section 4 connects to: Section 2 (receives signals and RDS band), Section 3 (parallel execution — neither determines the other), Section 6 (passes KPI movement package for causal explanation generation), Section 7 (passes display KPI set for response assembly).